

كلية الطب – جامعة العميد

الحملة التوعوية حول الحصول على

الاعتمادية

الرؤية

التميز الفعال من اجل رفد المجتمع بأطباء اكفاء في الخدمة الطبية والبحث العلمي

الرسالة

توفير بيئة جامعية تعمل بمنهج تعليمي مواكب للتطور
اطباء ذوي كفاءة ومستوى علمي قادرين على تقديم الخدمة المتميزة



الاهداف

- اعداد اطباء بكفاءة عالية
- التأكيد على مبادئ الاخلاق الطبية
- نشر مفاهيم العمل كفريق واحد
- العمل على حل المشكلات الصحية
- تطبيق مخرجات البحوث المتقدمة
- تعزيز عملية البحث العلمي
- تبني مفاهيم التعلم المهني المستمر



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Introduction- Metabolism and Elimination

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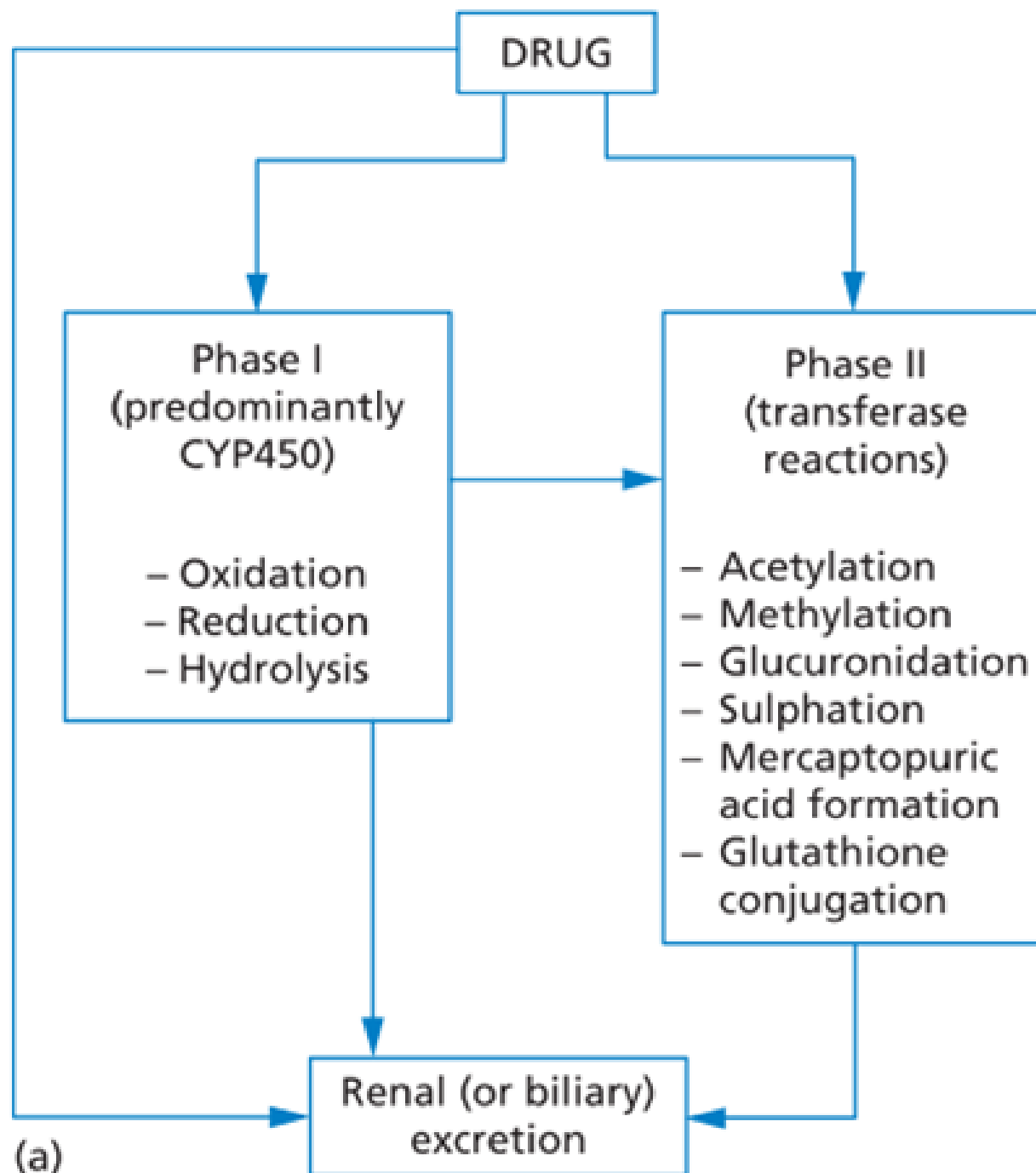
DRUG CLEARANCE THROUGH

METABOLISM:

- The liver is the most important site of drug metabolism (mostly by cytochrome P450 (CYP450) enzyme superfamily).
- *The drug metabolism is important due to:*
 - ✦ One of the primary mechanisms by which drugs are inactivated (like ethanol)
 - ✦ Some of drug activity is increased by metabolism, as activation of pro-drugs (e.g. hydrolysis of enalapril to enalaprilat).
 - ✦ The formation of polar metabolites from a non-polar drug permits efficient urinary excretion.

Pro-drugs

- Some enzymatic conversions yield active compounds with a longer half-life than the parent drug (e.g. diazepam has a half-life of 20–50 hours, whereas its pharmacologically active metabolite desmethyldiazepam has a plasma half-life of approximately 100 hours).



Phases of metabolism

A. Phase I reactions: involve a metabolic modification of the drug by introducing a polar functional group, such as -OH or -NH_2 . (commonly oxidation, reduction or hydrolysis).

Products of phase I are either pharmacologically active or inactive.

Phase I reactions utilizing the P_{450} system

- ▶ There are many different P_{450} isoforms.
- ▶ P_{450} enzymes exhibit considerable genetic variability among individuals and racial groups.

ENZYME INDUCTION:

- It is a process by which enzyme activity is enhanced, usually because of increased enzyme synthesis (e.g. nicotine, ethanol, rifampicin, carbamazepine, ...etc are enzyme inducers) leading to:
 - ❑ Increase activity of pro-drugs.
 - ❑ Decrease the life span of most of drugs.

ENZYME INHIBITION:

- Allopurinol, cimetidine, fluconazole, voriconazole, ritonavir, methotrexate, angiotensin converting enzyme inhibitors, non-steroidal anti-inflammatory drugs and many others have enzyme inhibition effect which can lead to:

- ☐ Drug accumulation and toxicity i.e. increase drug half life and plasma concentration.
- ☐ Reduce the efficacy of pro-drugs and failure of therapy.

Phase I reactions not involving the P450 system:

- These include amine oxidation (for example, oxidation of catecholamines or histamine), alcohol dehydrogenation (for example, ethanol oxidation), esterases (for example, metabolism of pravastatin in liver), and hydrolysis (for example, of procaine).

B. Phase II reactions are synthetic conjugation reactions. Phase II metabolites have increased polarity and are more readily excreted in the urine (or, less often, in the bile), and they are usually – but not always – pharmacologically inactive.

C. Reversal of order of the phases: Not all drugs undergo phase I then phase II reactions in that order e.g., isoniazid is first acetylated (a Phase II reaction) and then hydrolyzed to isonicotinic acid (a Phase I reaction).

Kinetics of metabolism:

1. First-order kinetics (linear kinetics):

- The metabolic transformation of drugs obeys Michaelis-Menten kinetics.

$$v = \text{rate of drug metabolism} = \frac{V_{\max} [C]}{K_m}$$

- That is, the rate of drug metabolism and elimination is directly proportional to the concentration of free drug i.e. it is not saturable.

Kinetics of metabolism:

1. Zero-order kinetics (non-linear kinetics):

- With a few drugs, such as aspirin, ethanol, and phenytoin, the doses are very large. Therefore $[C]$ is much greater than K_m , and the velocity equation becomes:

$$v = \text{rate of drug metabolism} = \frac{V_{\max} [C]}{[C]} = V_{\max}$$

- The enzyme is saturated by a high free-drug concentration and the rate of metabolism remains constant over time. A constant amount of drug is metabolized per unit of time, and the rate of elimination is constant and does not depend on the drug concentration.

ELIMINATION OF DRUGS

- ✓ Elimination of drugs requires the agents to be polar for efficient excretion.
- ✓ Removal of a drug from the body occurs via a number of routes, the most important being through the kidney into the urine.

Renal elimination of a drug

1. Glomerular filtration: Free drug (not bound to albumin) flows through the capillary slits into Bowman's space.

2. Proximal tubular secretion:

- ❖ Primarily occurs in the proximal tubules by two energy-requiring active transport systems: one for anions and one for cations.
- ❖ Thus, competition between drugs for these carriers can occur within each transport system (for example, probenecid with penicillin).

Renal elimination of a drug

3. Distal tubular reabsorption:

- ❖ As a drug moves toward the distal convoluted tubule, its concentration increases and exceeds that of the perivascular space.
- ❖ The drug, if uncharged, may diffuse out of the nephric lumen, back into the systemic circulation.

- Changing in pH of the urine to increase the ionized form of the drug in the lumen may increase clearance of undesired drug.

“Ion trapping.”

- Weak acids can be eliminated by alkalinization of the urine (phenobarbital overdose by bicarbonate),
- Elimination of weak bases may be increased by acidification of the urine (amphetamine by NH_4Cl).

Other routes of elimination

- like intestines, the bile, the lungs (halothane & isoflurane anesthetics), and milk in nursing mothers.
- Sweat, saliva, tears, hair, & skin drug excretion occurs to a small extent.

Important concepts

- Half-Life ($t_{1/2}$):

It is the time required to change the amount of drug in the body by one-half.

- Steady state:

It is a state at which *drug intake = drug clearance*.

Maintenance dose of a given drug is required to preserve steady state

Important concepts

Creatinine clearance:

- ❖ Creatinine is an end-product of muscle metabolism, a derivative of muscle creatine phosphate. It is produced continuously in the body and is excreted in the urine.
- ❖ It is useful to assess renal function in regard its ability to eliminate the drug (i.e. the dose of the drug must be adjusted if C_{cr} reduce)

$$GFR \approx C_{Cr} = \frac{U_{Cr} \times V}{P_{Cr}}$$

- Normal creatinine clearance for healthy men is 97-137 mL/min.
- Normal creatinine clearance for healthy women is 88-128 mL/min.